

Unraveling the semi-interpenetration with conducting polymer of electroresponsive click-hydrogels made of hyaluronic acid via molecular dynamics simulations

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ABSTRACT

Electroresponsive hydrogels have been prepared by semi-interpenetrating a hyaluronic acid-based click-hydrogel (clickHA) with different concentrations of poly(hydroxymethyl-3,4-ethylenedioxythiophene) (PEDOT-MeOH). The morphology of the resulting hydrogels, which showed excellent electrochemical response, consisted in a homogeneous distribution of PEDOT-MeOH nanoparticles decorating the pore well of the honeycomb-like clickHA hydrogel. In order to investigate this particular organization, atomistic molecular dynamics simulations have been conducted on clickHA hydrogels with and without hydroxymethyl-3,4-ethylenedioxythiophene (EDOT-MeOH) monomers. Results have shown that, while EDOT-MeOH molecules do not affect the physical and structural characteristics of clickHA, infiltrated monomers rapidly tend to aggregate inside the hydrogel, forming small clusters stabilized by π - π stacking interactions. These clusters probably act as polymerization nuclei for the formation of PEDOT-MeOH nanoparticles, which explains not only the morphology observed but also the remarkable electrochemical response of clickHA/PEDOT-MeOH hydrogels.

1. Introduction

Hydrogels are binary systems formed by a large amount of water held inside a three dimensional polymer network [1–6]. In this mesh, natural or synthetic polymer chains are covalently or physically networked using a crosslinker, making the system insoluble. In addition to water absorption capacity, hydrogels display many other advantageous properties, such as viscoelasticity, *softness*, *toughness*, *stretchability*, *deformability*, and *stimuli-responsiveness*, that can be easily tuned [1–7]. Although not all hydrogels have good biodegradability and biocompatibility, such properties can be easily achieved using natural polymers or adjusted by modifying the functional groups of synthetic polymers.

Hyaluronic acid (HA) is an important component of the extracellular matrix (ECM). This biodegradable non-branched polysaccharide is constituted by *N*-acetyl-D-glucosamine and D-glucuronic acid repeatedly

linked via glycoside bonds in the arrangement of alternating β -(1 $\rightarrow$ 4) and β -(1 $\rightarrow$ 3) bonds [8]. HA hydrogels are very attractive as biomaterials for biomedical applications due to their water retention capacity, biocompatibility, and versatility for modification [9]. The latter has been extensively used to improve the stability and mechanical strength of HA hydrogels, while their biocompatibility and biodegradability have been retained [10,11]. Because of its abundance in the dermis and epidermis (*i.e.* 50 % of total HA is localized in skin) [11–13], HA hydrogels have been widely employed for wound care and skin tissue regeneration [14–19], as well as skin repairmen and rejuvenation (*i.e.* cosmetics) [20–22].

In a recent study, we prepared an electroresponsive HA-based click-hydrogel by semi-interpenetrating a conducting polymer (CP), named poly(hydroxymethyl-3,4-ethylenedioxythiophene) (PEDOT-MeOH), into a modified HA hydrogel [23]. Specifically, HA was functionalized

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with thiol moieties (HA-SH) and, subsequently, crosslinked with an alkyne-modified 3-arm polyethylene glycol precursor (PEG) [24]. The resulting hydrogel, named clickHA (Scheme 1), was infiltrated with water soluble hydroxymethyl-3,4-ethylenedioxythiophene (EDOT-MeOH) monomer, which was subsequently polymerized inside the hydrogel by chemical oxidation using ammonium persulfate (APS), as sketched in Scheme 2. The obtained conducting hydrogel, clickHA/PEDOT-MeOH, showed a homogeneous distribution of PEDOT-MeOH inside the clickHA matrix, which induced significant electrochemical activity and reflected the compatibility of the two components. In addition, clickHA/PEDOT-MeOH was found to be biocompatible, promoting cells mobility when used as electrical contact upon electrostimulation with a small voltage (+0.5 V) for a short time (15 min) [23]. Indeed, *in vitro* wound healing assays proved that, under such conditions, wound closure occurred in 1 h only.

In this work, we go one step further by analyzing the molecular interactions that led the formation of semi-interpenetrated clickHA/PEDOT-MeOH. For this purpose, we have initially examined the effect of EDOT-MeOH concentration infiltrated into the clickHA hydrogel to form clickHA/PEDOT-MeOH. More specifically, the internal morphology and electrochemical properties of clickHA/PEDOT-MeOH prepared using different EDOT-MeOH concentrations (*i.e.* 0.02, 0.05 and 0.15 M) have been studied. After that, atomistic molecular dynamics (MD) simulations have been conducted considering clickHA hydrogels without and with infiltrated EDOT-MeOH monomers, the latter being denoted clickHA/EDOT-MeOH. MD simulations have provided insights into the molecular properties of clickHA and clickHA/EDOT-MeOH hydrogels, while the analysis of the interactions between EDOT-MeOH monomers have allowed us to understand the particular organization of the CP in clickHA/PEDOT-MeOH hydrogels.

2. Methods

2.1. Synthesis of clickHA hydrogels

The synthesis of clickHA (HA-S-ene-PEG) hydrogels from HA-SH [25, 26] and 3-arm alkyne-PEG [24] (Scheme 1) was reported in our previous study [23]. Briefly, 3-arm alkyne-PEG (3.5 mg) dissolved in phosphate buffered saline (PBS) solution (100 μ L) was added to a HA-SH (31.3 mg) solution in PBS (300 μ L) at pH 7.4 (adjusted with KOH 1 M). The mixture was vigorously stirred for 30 s and left overnight at 4 °C. The resulting hydrogels were washed twice with water.

2.2. Synthesis of semi-interpenetrated clickHA/PEDOT-MeOH hydrogel

The clickHA preformed hydrogels were immersed at room temperature in 10 mL of a 0.02, 0.05 or 0.15 M of EDOT-MeOH (95 %) monomer solution with 0.2 M HCl overnight. Then, the chemical oxidative polymerization was conducted by immersing the infiltrated

hydrogels in 10 mL of a 0.1 M APS (≥ 98 %) solution. The resulting semi-interpenetrated hydrogels were thoroughly washed (1 h) and kept moist at 4 °C for characterization tests. For electrochemical assays, the semi-interpenetrated hydrogels were neutralized by immersing in NaOH 0.2 M and, subsequently, washed with Milli-Q® water until the rinse water had neutral pH. Depending on the concentration of EDOT-MeOH used in the infiltration process, which could be 0.02, 0.05 or 0.15 M, the obtained hydrogels have been denoted clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) or clickHA/PEDOT-MeOH(0.15), respectively.

2.3. Characterization

FTIR spectra were recorded with a Jasco 4100 spectrophotometer. Samples were placed on an attenuated total reflection accessory (ATR) with a diamond crystal. For each sample, 16 scans with a resolution of 4 cm^{-1} were taken between 4000 and 600 cm^{-1} . For hydrogels, samples were lyophilized previously for 24 h.

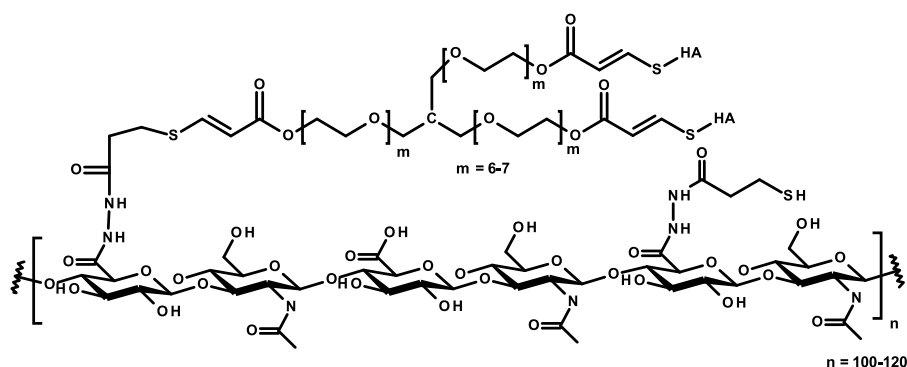
To characterize morphological features, a scanning electron microscopy (SEM) was used and operated at 5 kV (Zeiss Neon40 outfitted with a focused ion beam with an energy-dispersive X-ray (EDX) spectroscopy system). To prevent electron charging issues, all samples were sputter-coated with a thin carbon layer using a K950X Turbo Evaporator. Prior to SEM observation, samples were lyophilized by freeze-drying. Nanoparticles size was determined from SEM micrographs using the software ImageJ (Wayne Rasband and contributors, National Institutes of Health, USA).

The electrochemical response of the prepared hydrogels was evaluated using cyclic voltammetry (CV). For this purpose, cylindrical samples (diameter: 9 mm; height: 6 mm) were shaped in a mold. CV studies were conducted with a potentiostat VIONIC (Metrohm Autolab B.V., Utrecht, The Netherlands) connected to a conductivity meter cell by means of two stainless steel electrodes AISI 304 isolated by a resin holder [27], samples being placed between the two electrodes. CV measurements ($n = 4$) were performed using the potential range from -0.2 to $+0.8$ V at a scan rate of 50 mV/s.

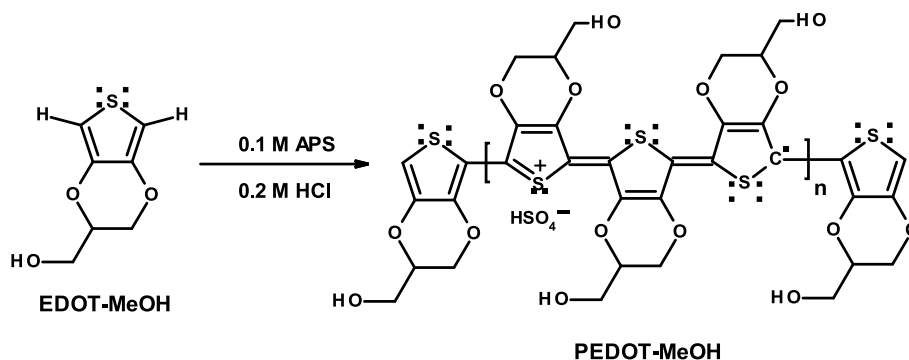
Electrochemical impedance spectroscopy (EIS) diagrams were taken at open circuit (OCP) over the frequency range of 100 kHz to 10 Hz with potential amplitude of 0.05 V using a VIONIC potentiostat/galvanostat.

2.4. Computational studies

Molecular models. The solid structure of HA was initially constructed using the in-house program SuSi [28]. The final model comprised 12 polymer chains, each consisting of 90 identical repeating units of HA. The structure was encapsulated in a cubic box with an edge length of 110 Å. This amorphous organization served as the foundation for creating hydrogel crosslinks. In the final hydrogel structure, 37 % of the carboxyl groups were transformed into hydrazine alkyl thiol.



Scheme 1. Chemical structure of clickHA.



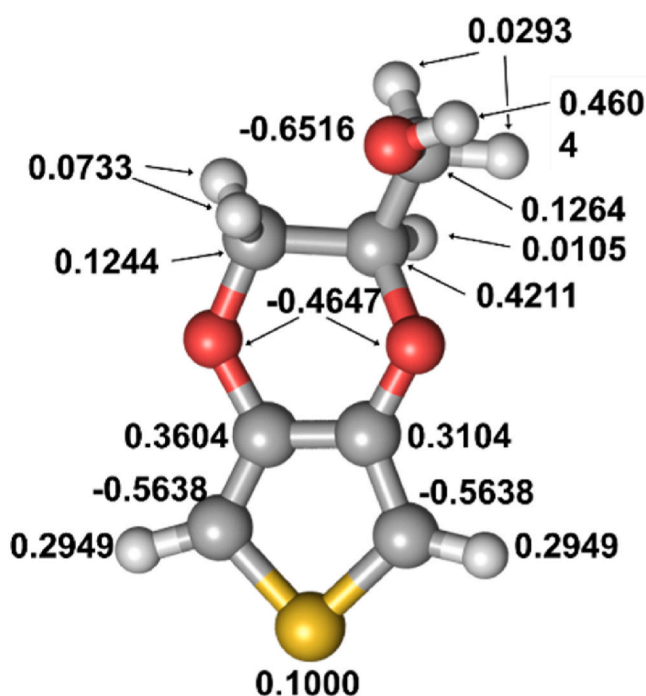
Scheme 2. Chemical oxidative polymerization of EDOT-MeOH to produce PEDOT-MeOH.

Subsequently, 10 % of the thiol-modified repeat units were crosslinked using the assigned crosslinker, resulting in an average of 33 hydrazine-modified carboxylic units and 3 crosslinks per HA polymer chain. All modifications to the HA system to obtain the final hydrogel were performed using Polymatic [29], a set of codes for the structure generation of polymers, utilizing an algorithm that simulates polymerization. The relaxation of the crosslinked structure was conducted using the LAMMPS program [30] (details can be found in the Electronic Supporting Information).

This crosslinked 3D system was used as the initial scaffold to build the fully hydrated hydrogel under low pH conditions (*i.e.* with all the acid groups in their protonated state), which are the conditions experimentally used for the *in situ* EDOT-MeOH (*i.e.* inside the hydrogel bulk). Before initiating the production simulations, adjustments were made to ensure adequate ionic strength, charge neutrality, and the proper EDOT-MeOH molarity to replicate the experimental polymerization conditions. Specifically, for the clickHA hydrogel model, 848 particles of each ion were introduced. Then, the molecular model was submerged into a previously NPT-equilibrated cubic box (with an edge close to 200 Å) filled with TIP3P [31] model water molecules. The final clickHA model comprised 517523 explicit atoms. For the clickHA/EDOT-MeOH model, the crosslinked 3D system was supplemented with the necessary chloride ions, sodium ions, and EDOT-MeOH monomers. Specifically, 848 particles of each ion were introduced, along with 165 EDOT-MeOH molecules. Finally, the molecular model was submerged into the previously NPT-equilibrated cubic box filled with TIP3P model [31] water molecules. All water molecules that overlapped with any atom of the solutes or extended beyond the box size of the solid polymer were removed. The final molecular model for clickHA/EDOT-MeOH comprised 505170 explicit atoms.

clickHA and clickHA/EDOT-MeOH hydrogels MD simulations. All intra- and intermolecular interactions were computed using the GAFF AMBER force-field [32], except for the electrostatic parameters. Force-field atom types for EDOT-MeOH are shown in Scheme S1, while stretching, bending, torsional (both proper and improper) and Lennard-Jones parameters are listed in Tables S1–S5. Atomic point charges were derived from the ESP charges of several model molecules, which were calculated using quantum mechanics. For each case, the most relevant conformations were used, and each set of point charges derived from each conformation was weighted by a Boltzmann factor. Atomic charges for clickHA were previously reported [23], while Scheme 3 shows the atomic point charges developed to model EDOT-MeOH.

Simulations were conducted using the GROMACS molecular package, version 2023 [33]. The time step was set at 2 fs, and bond lengths involving hydrogen atoms were constrained using the LINCS algorithm [34]. Atom pair distance cutoffs were applied at 14 Å to compute all van der Waals interactions. To avoid discontinuities in this energy component, the van der Waals energy term was forced to converge to zero gradually by applying a smoothing factor from 10 Å. Electrostatic interactions were extensively computed using Ewald summations. The real



Scheme 3. Atomic point charges used to model the electrostatic interactions involving EDOT-MeOH monomers.

space term was defined by the van der Waals cutoff (14 Å), while the reciprocal space was computed by interpolating the effective charge into a charge mesh with a grid thickness of 1 point per Å³ (particle mesh Ewald) [35].

Before production MD runs, a series of equilibration runs were performed: 3000 steps of energy minimization, 0.5 ns of isothermal equilibration at 298 K (NVT conditions) using the Berendsen thermostat [36], followed by 1 ns of isobaric equilibration (NPT conditions) at the same temperature and 1.013 bar using the extended-ensemble Parrinello-Rahman thermobarostat [37,38]. The last snapshot of that final run served as the starting point for a 50 ns simulation under the same conditions as the final equilibration run.

3. Results and discussion

3.1. Semi-interpenetration of PEDOT-MeOH into clickHA hydrogel

ClickHA (control), clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) hydrogels were prepared using the procedures described in the Methods section. The key-steps of such strategy were: 1) preformed clickHA hydrogels were

immersed in a 0.02, 0.05 or 0.15 M of EDOT-MeOH solution overnight, favoring the diffusion of the monomer through the hydrogel network; 2) PEDOT-MeOH was formed by chemical oxidative polymerization immersing the infiltrated hydrogels in APS. It is worth mentioning that the limit of solubility of EDOT-MeOH in water is slightly below 0.15 M, which ensures that clickHA/PEDOT-MeOH(0.15) incorporates the maximum amount of CP possible.

As expected, the amount of incorporated CP affected the transparent clickHA hydrogel, as is reflected by the digital photographs shown in Fig. 1a. Thus, the clickHA/PEDOT-MeOH(0.02) hydrogel was green, while clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) were black. Fig. 1b compares the FTIR spectra of lyophilized clickHA (control), clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15). The FTIR spectrum of clickHA hydrogel displays absorption peaks corresponding to the stretching of the C–O–C group (1027 cm^{-1}), asymmetric C=O and symmetric C–O stretching modes of the COO^- groups (1610 and 1564 cm^{-1} , respectively) [39]. The spectra acquired for clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) exhibited not only the characteristic fingerprints associated with clickHA, but also some bands attributed to PEDOT-MeOH. More specifically, the bending of the C–S–C (684 cm^{-1}), C=C asymmetric stretching (1152 cm^{-1}), C–O–C bending vibration in ethylenedioxy groups (1246 cm^{-1}), and C–C inter-thiophene ring stretching (1471 cm^{-1}) [40]. In addition, the C=O stretching observed at 1610 cm^{-1} for clickHA, which was obtained at slightly basic pH, changed to $1650\text{--}1697\text{ cm}^{-1}$ for the three CP-containing hydrogels since EDOT-MeOH was polymerized in acid medium.

Fig. 1c compares the cyclic voltammograms recorded for clickHA hydrogels without and with PEDOT-MeOH. The control does not show electrochemical activity in comparison to the three PEDOT-MeOH containing hydrogels, evidencing the significant role of the CP in the electrochemical response. As expected, the voltammetric area increases with the content of PEDOT-MeOH, even though such increment is much higher when clickHA/PEDOT-MeOH(0.02) is compared with clickHA/PEDOT-MeOH(0.05) than when the latter is compared with clickHA/PEDOT-MeOH(0.15). More specifically, the electroactivity, which corresponds to the area of the anodic and cathodic scans, is lower for clickHA/PEDOT-MeOH(0.02) than for clickHA/PEDOT-MeOH(0.05) by a factor of 3.2, whereas it is lower for clickHA/PEDOT-MeOH(0.05) than

for clickHA/PEDOT-MeOH(0.15) by a factor of 1.4. These observations suggest that percolation of PEDOT-MeOH chains is reached when the concentration of conducting monomer is $\geq 0.05\text{ M}$.

In order to quantify the effect of the amount of incorporated CPs on the electrochemical activity, the area of the voltammograms was determined. We found that the voltammetric charge of clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) was 3.14 and 4.64 times higher than that of clickHA/PEDOT-MeOH(0.02), respectively, while the voltammetric charge of clickHA/PEDOT-MeOH(0.15) was only 1.48 times higher than that of clickHA/PEDOT-MeOH(0.05). These results indicated that changing the EDOT-MeOH concentration from 0.02 M to 0.05 M (i.e. $\times 2.5$) resulted in a large improvement in electrochemical activity, while increasing it to 0.15 M (i.e. $\times 7.5$) resulted in only a small improvement over 0.05 M.

On the other hand, it is worth noting that the increment in the area with respect to the control clickHA hydrogel is much higher than that achieved for other semi-interpenetrated hydrogels, in which the CP was incorporated inside an ionic hydrogel using the same polymerization technique [41], (i.e. chemical polymerization) or even more efficient strategies (e.g. incorporating CP nanoparticles [42] alone or combined with electrochemical polymerization processes [43]).

SEM micrographs of the cross-section reflect the highly porous structure of clickHA (Fig. 2a) with interconnected pores of $7.1 \pm 1.6\text{ }\mu\text{m}$ ($n = 50$). The organization of the pores, which defines a honeycomb-like morphology, is expected to facilitate the transport of ions toward the inner site of the hydrogels, explaining the electrochemical response of clickHA. Besides, clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) preserve the honeycomb-like structure (Fig. 2b–d, left), even though the shape of the pores is more irregular for the CP-containing hydrogels than for the control one. Furthermore, for clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15), the pore walls were extensively decorated with PEDOT-MeOH nanoparticles (Fig. 2b–d, right), suggesting the formation of an electronically conductive percolated network. Indeed, both the amount and size of PEDOT-MeOH nanoparticles, which increased with the concentration of conducting monomer in the infiltration step, is fully consistent with the enhancement of the electrochemical response, as observed by CV (Fig. 1c).

The size of PEDOT-MeOH nanoparticles was 2.90 and 4.55 times

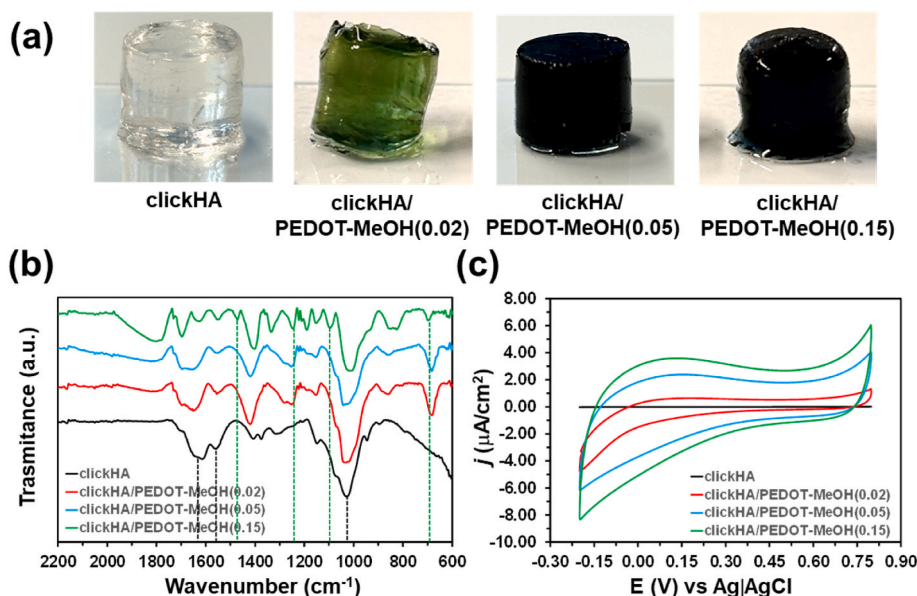


Fig. 1. (a) Digital photographs, (b) FTIR spectra and (c) cyclic voltammograms of clickHA, clickHA/PEDOT-MeOH(0.02), clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15) hydrogels.

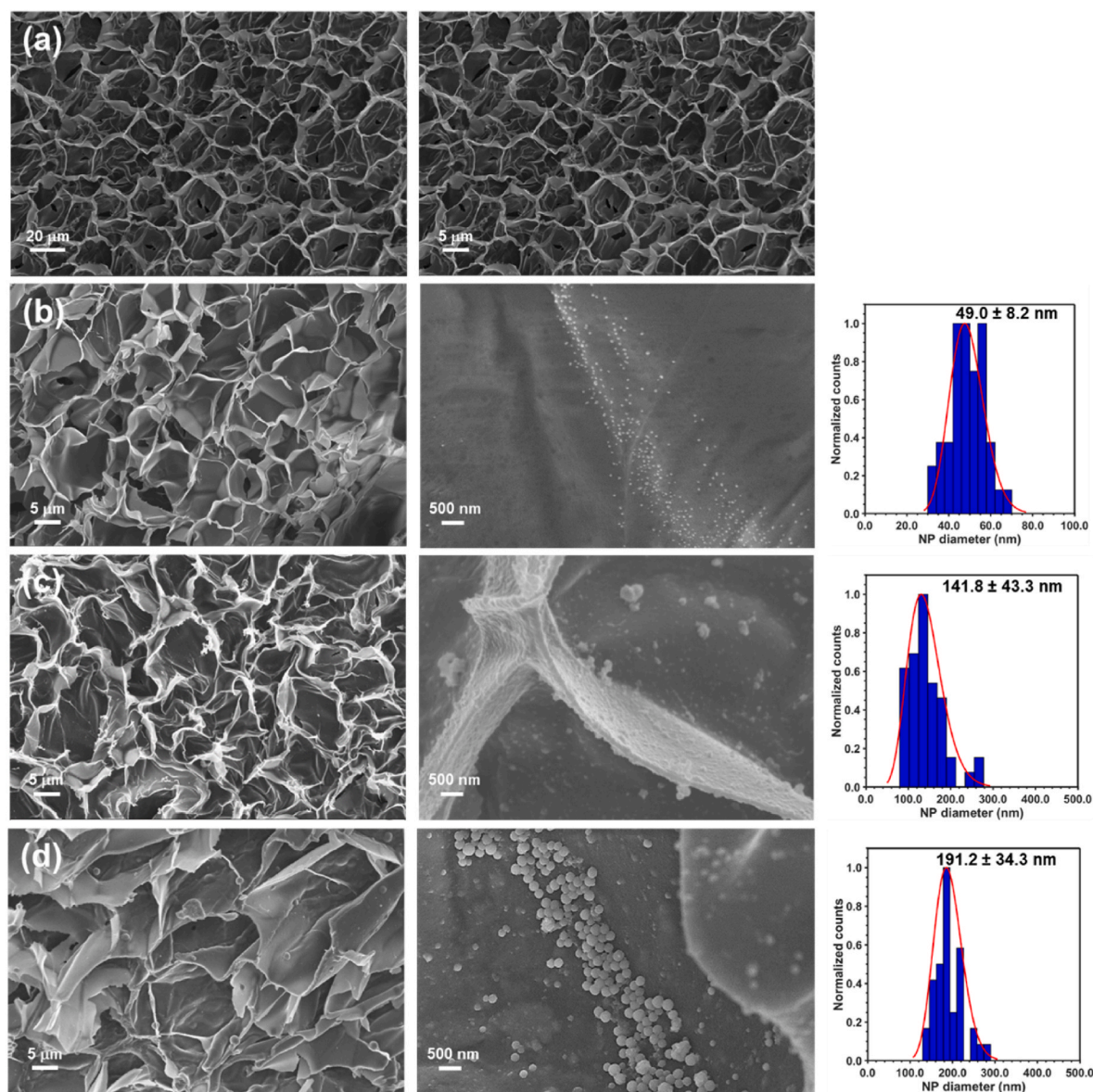


Fig. 2. SEM micrographs of the cross-section of (a) clickHA, (b) clickHA/PEDOT-MeOH(0.02), (c) clickHA/PEDOT-MeOH(0.05) and (d) clickHA/PEDOT-MeOH(0.15) hydrogels.

smaller for clickHA/PEDOT-MeOH(0.02) than for clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15), respectively, while the size was 1.34 times higher for clickHA/PEDOT-MeOH(0.15) than for clickHA/PEDOT-MeOH(0.05). These values are in excellent agreement with the relative increase for electrochemical activity discussed above (*i. e.* 3.14 and 4.64 times higher for clickHA/PEDOT-MeOH(0.05) and clickHA/PEDOT-MeOH(0.15), respectively, than for clickHA/PEDOT-MeOH(0.02)). According to these results, a significant improvement in the charge transport efficiency is obtained by changing the EDOT-MeOH monomer concentration from 0.02 M to 0.05 M, while the improvement is residual when it is increased to 0.15 M. This indicates that a concentration of 0.05 M is enough to achieve the percolation between the PEDOT-MeOH chains. This has been attributed to the fact that the diffusion of monomer molecules within the clickHA hydrogel is not favored, possibly due to a gradient effect, when the monomer concentration is excessive.

Specifically, clickHA/PEDOT-MeOH(0.02) showed very small PEDOT-MeOH nanoparticles with an average size of 49 ± 8 nm, as determined from the corresponding size distribution curve, that were dispersed on the walls of the clickHA hydrogel (Fig. 2b). The size of the

nanoparticles increased to 142 ± 43 nm for clickHA/PEDOT-MeOH(0.05). Furthermore, in this case, CP nanoparticles organized in percolated structures forming conduction paths through the hydrogel matrix (Fig. 2c), which explains the significant enhancement of the electrochemical response with respect to clickHA/PEDOT-MeOH(0.02). Finally, for clickHA/PEDOT-MeOH(0.15), the PEDOT-MeOH nanoparticles showed an average size of 191 ± 34 nm and grouped defining very well-defined conduction paths (Fig. 2d).

The organization of the PEDOT-MeOH molecules forming a network of percolated nanoparticles that decorate the walls of the clickHA pores, instead of being structured to form a homogeneous layer that covers the hydrogel structure, as is usual, suggests that the EDOT-MeOH molecules infiltrated in the hydrogel tend to group together, facilitating the formation of said particles during polymerization. In order to determine whether this hypothesis is correct, the following sections present the results obtained from MD simulations using a model system formed by a clickHA hydrogel and EDOT-MeOH molecules infiltrated into its pores. Since the electrochemical response obtained for the clickHA/PEDOT-MeOH(0.05) hydrogel is comparable to that of clickHA/PEDOT-MeOH(0.15), the simulated system corresponds to the clickHA hydrogel

infiltrated with a 0.05 M EDOT-MeOH solution. For simplicity, hereafter, the simulated model system is denoted clickHA/EDOT-MeOH, without making reference to the amount of monomer contained inside the hydrogel.

The electrical resistances (R) of clickHA and clickHA/PEDOT-MeOH (0.05), as determined by EIS (Fig. S1), were 7.6 ± 0.8 - and 2.3 ± 0.5 -k Ω /cm², respectively, supporting the electron transport in the latter. Table 1 compares the resistance of clickHA/PEDOT-MeOH(0.05), as well as the previously determined Young's modulus (E) [23], with those reported for some PEDOT-containing biohydrogels [44–46]. Alginate/PEDOT hydrogels [44] exhibit higher resistivity and rigidity than clickHA/PEDOT-MeOH, which explains their utilization in technological applications, such as pressure sensors. On the other hand, the resistivity poly(γ -glutamic acid)/PEDOT [45] and cellulose/PEDOT/Al₂O₃ [46] is comparable and lower, respectively, than that of clickHA/PEDOT-MeOH. Unfortunately, the mechanical properties of the two formers hydrogels were not determined, as they were used for energy related applications.

3.2. Modeling the effect of monomer on the internal structure of the clickHA hydrogel matrix

Polymer simulations provide explanations of experimentally observed macromolecular as has been reported [47,48]. In this work, MD simulations were performed to gain understanding of the microstructure observed for clickHA/PEDOT-MeOH hydrogels in the previous section. To this end, the effect of EDOT-MeOH monomer incorporation on the internal structure of the clickHA hydrogel matrix was initially examined by analyzing the MD trajectories obtained for clickHA and clickHA/EDOT-MeOH model systems. It should be mentioned that the computational expense of explicitly representing every atom of the system limits atomistic MD simulations to a timescale of tens of nanoseconds. Despite this limitation, MD simulations are very useful to describe the events occurring in short time-length scales, as for example the aggregation EDOT molecules studied in this work.

The average density of clickHA and clickHA/EDOT-MeOH, which remained practically constant during the production run (Fig. 3a), was 1.018 ± 0.001 and 1.060 ± 0.001 g/cm³, respectively. The 4 % higher density of clickHA/EDOT-MeOH compared to clickHA was attributed to the slightly greater weight of the EDOT-MeOH monomer compared to the water molecules that must move to occupy its volume within the clickHA hydrogel. The good accommodation of the EDOT-MeOH monomer to the free space of the clickHA matrix was evidenced by the fact that the unoccupied volume was practically identical for clickHA and clickHA/EDOT-MeOH (i.e. $86.8 \% \pm 0.1 \%$ and $86.4 \% \pm 0.1 \%$, respectively). Moreover, the unoccupied volume, which is defined as the inverse of the van der Waals volume of the hydrogel matrix atoms in the generated models (i.e. excluding water molecules, ions, and EDOT-MeOH monomers), remained stable throughout the whole MD trajectory (Fig. 3b). This result suggests that neither the incorporation of EDOT-OH molecules nor the relaxation of their positions produced deformations in the hydrogel matrix, allowing us to hypothesize that the infiltration of the conducting monomer does not significantly affect the clickHA internal structure despite the 4 % increase in density.

The solvent accessible surface area (SASA) also supported the

hypothesis about the negligible influence of EDOT-MeOH monomer accommodation on the clickHA structure. Thus, the temporal evolution of the SASA for clickHA and clickHA/EDOT-MeOH model systems showed an attenuated reduction up to ~ 15 ns, followed by a stabilization (Fig. 3c). The SASA averaged over the last 25 ns of trajectory, which was evaluated considering only the hydrogel matrix, was 2130 ± 200 nm² and 2220 ± 170 nm² for clickHA and clickHA/EDOT-MeOH, respectively. Accordingly, the incorporation of EDOT-MeOH resulted in an expansion of the structure of 4 % only, supporting that the impact of the monomer in the hydrogel matrix is minimum.

The pore size in the simulated models was estimated by computing the largest cavity diameter (LCD) and the pore limiting diameter (PLD), which were calculated using the PoreBlazer v4.0 software [49]. The LCD corresponds to the maximum diameter of spherical probe particle that can fit within the hydrogel structure without overlapping any polymer atom, while the PDL is the largest probe that can cross the hydrogel defined within the simulation box in at least one dimension via a diffusive pathway, respectively. Analyses of the temporal evolution of LCD and PLC for clickHA and clickHA/EDOT-OH (Fig. 4a and b) corroborate that the impact of the EDOT-OH monomer in the hydrogel matrix is not significant. The average values, which stabilize after around 15–20 ns, considering the last 25 ns of trajectory are 113 ± 4 Å and 89 ± 4 Å for the LCD of clickHA and clickHA/EDOT-OH, respectively, and 50 ± 3 Å for the PLC of both clickHA and clickHA/EDOT-OH. Thus, although the accommodation of EDOT-OH molecules induces a reduction of around 21 % in the LCD, which is consistent with the increase of the density, the diffusion pathway remains unaltered, fact that proves that the average PLC is identical for the two systems. Fig. 4c graphically illustrates how EDOT-OH monomers do not alter appreciably the diffusion pathway and the internal structure of the hydrogel matrix. Thus, after remove all species different from the clickHA hydrogel, the superposition of two snapshots randomly taken from the production trajectories with and without EDOT-OH shows important similarities in the internal structure of the crosslinked hydrogel network.

3.3. Modeling the diffusion of the infiltrated monomer

The diffusion coefficients of water and EDOT-OH were calculated using the temporal evolution of the mean-squared displacement (Δr^2) of the oxygen atom of each water molecule and the center of mass of each monomer, respectively. At long times (t), the slope of the linear diffusive regime gives the translational self-diffusion coefficient (D_{wat} and D_{EDOT}) according to the Einstein relation:

$$D = \frac{1}{6} \frac{\delta(\Delta r^2)}{\delta t} \quad (1)$$

where D is the diffusion coefficient obtained in the simulation box using periodic boundary conditions.

The linearity of the Δr^2 for water is shown in Fig. 5a, the calculated diffusion coefficient being of $3.44 \cdot 10^{-9}$ m/s, which is in very good agreement with other values reported in the literature [50,51]. For EDOT-MeOH monomer, the variation of Δr^2 with the simulation time, which is plotted in Fig. 5b, exhibits three regions (A, B and C). The diffusion coefficient of EDOT-MeOH increases from $3.9 \cdot 10^{-12}$ m/s (A, from 0 to 30 ns) to $1.0 \cdot 10^{-11}$ m/s (B, from 30 to 42 ns) and $3.8 \cdot 10^{-11}$ m/s (C, above 42 ns). As the diffusion depends on the intermolecular interactions, these results indicate not only that the diffusion of water is much faster than that of EDOT-MeOH, as expected, but also that the relaxation is much slower for the interactions between EDOT-MeOH and the clickHA matrix than for water...clickHA interactions. Thus, much longer trajectories should be conducted to obtain a reliable value of the diffusion coefficient for EDOT-MeOH, which is out of the scope of this study that is focused on understanding the structure of the semi-interpenetrated PEDOT-MeOH.

Even though the diffusion coefficient of EDOT-MeOH is much lower

Table 1

Comparison of the resistivity and Young's modulus of different PEDOT-containing biohydrogels.

Hydrogel	R (k Ω /cm ²)	E (kPa)	Reference
clickHA/PEDOT-MeOH(0.05)	2.3 ± 0.5	29.7 ± 4.3	This work and [23]
Alginate/PEDOT	48.9 ± 9.8	~ 500	[44]
Poly(γ -glutamic acid)/PEDOT	6.38	–	[45]
Cellulose/PEDOT/Al ₂ O ₃	0.04	–	[46]

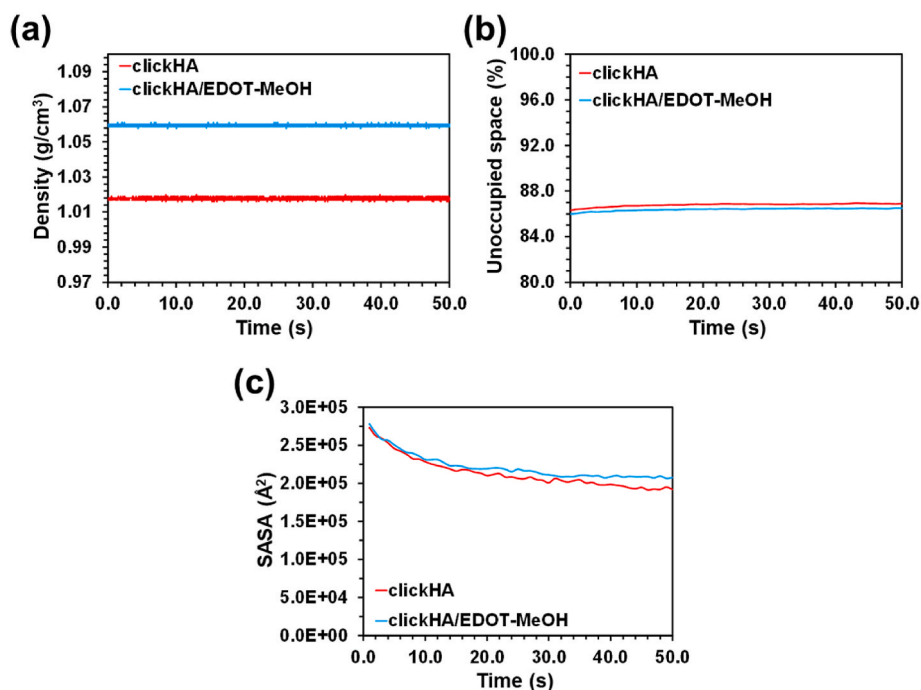


Fig. 3. Temporal evolution of the (a) density, (b) unoccupied space and (c) SASA for clickHA and clickHA/EDOT-MeOH.

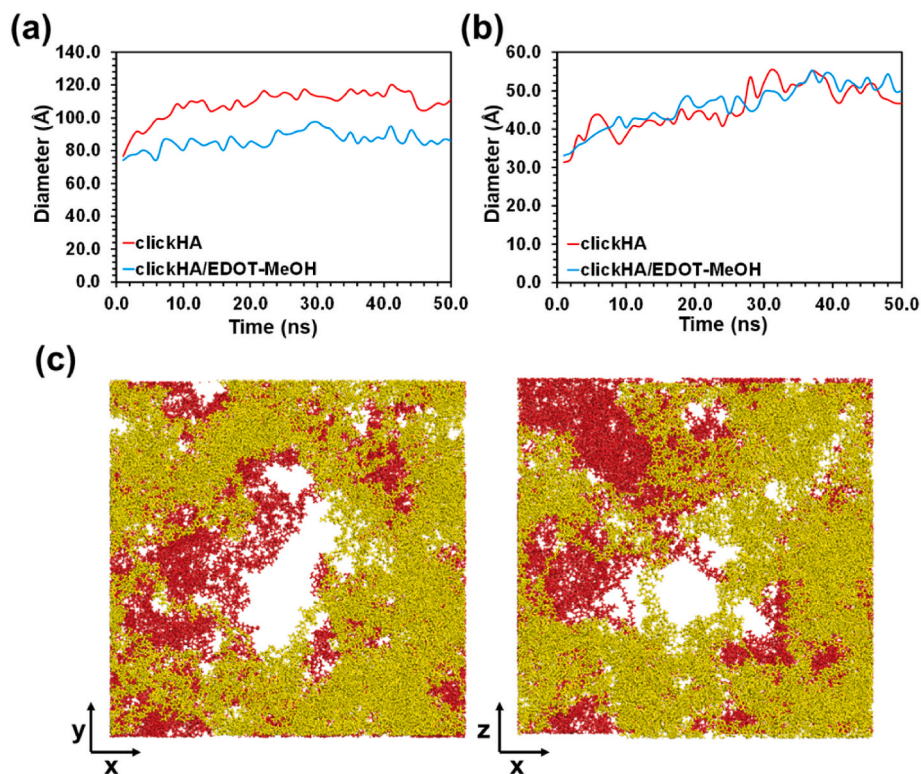


Fig. 4. Temporal evolution of the (a) LCD and (b) PLD for clickHA and clickHA/EDOT-OH. (c) Superposition of representative snapshots for both clickHA (yellow) and clickHA/EDOT-OH (red) using the corresponding equatorial (right panel) and axial (left panel) projections. All species (*i.e.* water molecules, ions and EDOT-OH monomers) have been removed, while HA polymer chains and PEG crosslinkers are represented by solid sticks. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

than that of water, the mobility of the monomer molecules is large enough to fully explore the simulation box during the 50 ns trajectory and, therefore, to group forming incipient clusters of monomers stabilized through π - π stacking and/or hydrogen bonding interactions. This

fact is reflected in Fig. 5c and d, which shows the position distribution probability of all EDOT-MeOH molecules and three randomly chosen EDOT-MeOH molecules through the MD production trajectory.

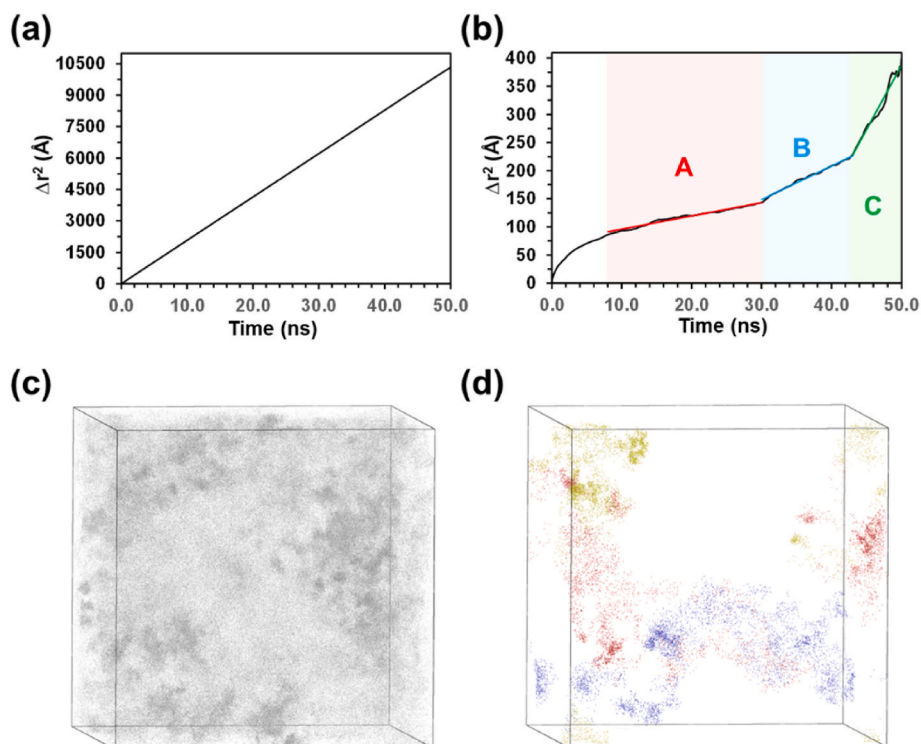


Fig. 5. Mean-squared displacement (Δr^2) for (a) water and (b) EDOT-MeOH molecules. The temporal Δr^2 evolution for EDOT-MeOH in (b) exhibits three different regimes, labeled as A, B and C, and marked with different background colors. (c, d) Position distribution probability of (c) all EDOT-MeOH molecules and (d) three randomly chosen EDOT-MeOH molecules through the MD production trajectory. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

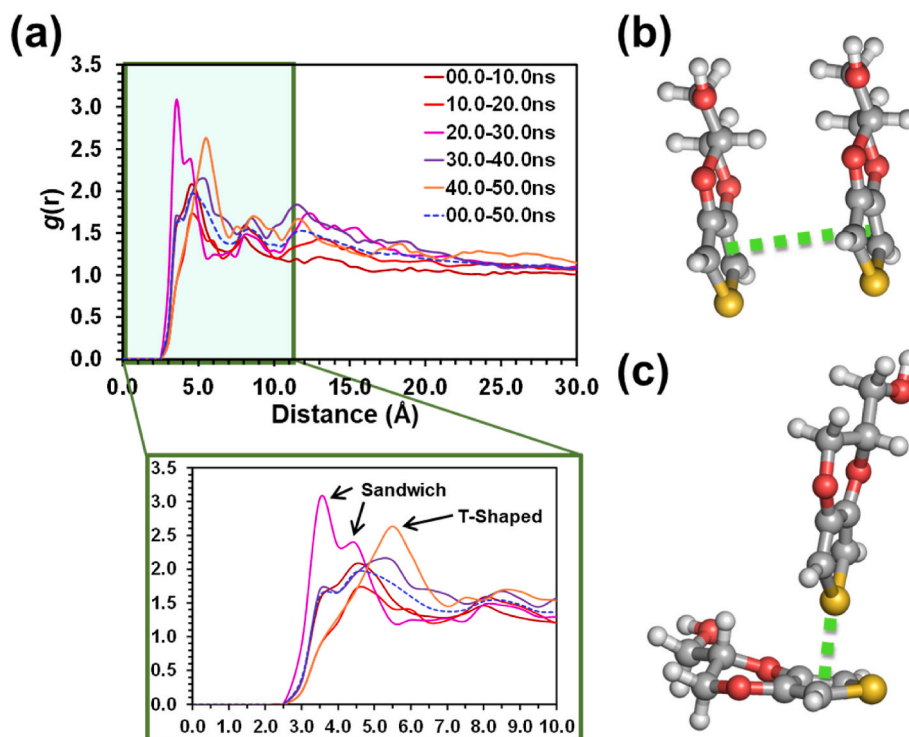


Fig. 6. Analyses of inter-monomer EDOT-MeOH...EDOT-MeOH π - π stacking interactions. (a) Radial distribution function, $g(r)$, of the center of masses of EDOT-MeOH...EDOT-MeOH pairs considering different intervals of time during the MD trajectory. (b, c) Sketches of the (b) sandwich-like and (c) T-shaped-like π - π stacking interactions between two EDOT-MeOH monomers.

3.4. Analyses of the EDOT-MeOH...EDOT-MeOH interactions

With the aim of explaining the organization of polymerized EDOT-MeOH monomers in a network of percolated PEDOT-MeOH nanoparticles, the possible formation of intermolecular EDOT-MeOH...EDOT-MeOH π - π stacking and hydrogen bonding interactions have been examined. For this purpose, the radial distribution function of the center of masses of the aromatic rings in EDOT-MeOH...EDOT-MeOH pairs, $g(r)$, was evaluated considering six intervals of time: 1) from 0 to 10 ns; 2) from 10 to 20 ns; 3) from 20 to 30 ns; 4) from 30 to 40 ns; 5) from 40 to 50 ns; and 6) the whole 50 ns of trajectory. The calculated $g(r)$, which are displayed in Fig. 6a, confirmed that the interactions associated with the monomer were still not relaxed, as the shape of the profiles changed with time.

The peaks at distances at around 3.5–4 Å are indicative of the planar (sandwich)-like π - π stacking interactions (Fig. 6b), while the peaks at around 5–5.5 Å can be associated with the T-shaped-like π - π stacking interactions (Fig. 6c) [52,53]. As it can be seen, the evolution of $g(r)$ suggests that π - π stacking with a sandwich-like configuration is slowly developing during the first 20 ns of the trajectory, consolidating in the interval from 20 to 30 ns. After this, the sandwich-like configuration gradually changes to a T-shaped-like configuration, as observed in the

profile obtained for the interval from 30 to 40 ns, which is consolidated in the last part of the trajectory, as shown for the $g(r)$ obtained from 40 to 50 ns.

Two types of intermolecular hydrogen bonds have been detected between EDOT-MeOH...EDOT-MeOH pairs: (i) O-H...O(H) between the hydroxyl groups of different monomers (green circle in Fig. 7a); and (ii) O-H...O(C) between the hydroxyl group of one monomer and the oxygen from the dioxane ring of another monomer (blue circle in Fig. 7a). Intermolecular O-H...O(H) and O-H...O(C) hydrogen bonds have been analyzed considering the $g(r)$ between H...O pairs involving the hydroxyl hydrogen atom of one monomer and the oxygen atom from the hydroxyl group or the dioxane ring of another monomer. In both cases, the $g(r)$ was calculated considering the six intervals of time disclosed above for π - π stacking interactions. Fig. 7b displays the profiles obtained for O-H...O(H) hydrogen bonds, which indicates that such interactions, which are identified by the peak at around 2 Å, are not particularly abundant. Indeed, the analysis of the individual temporal profiles show that O-H...O(H) interactions are not formed during the first 10 ns of trajectory. After that, in the next 20 ns, the interactions form and disrupt, remaining finally stable after 30 ns of trajectory. Overall, although O-H...O(H) hydrogen bonds are clearly detected through the 50 ns trajectory, they are not abundant enough to be the driving force

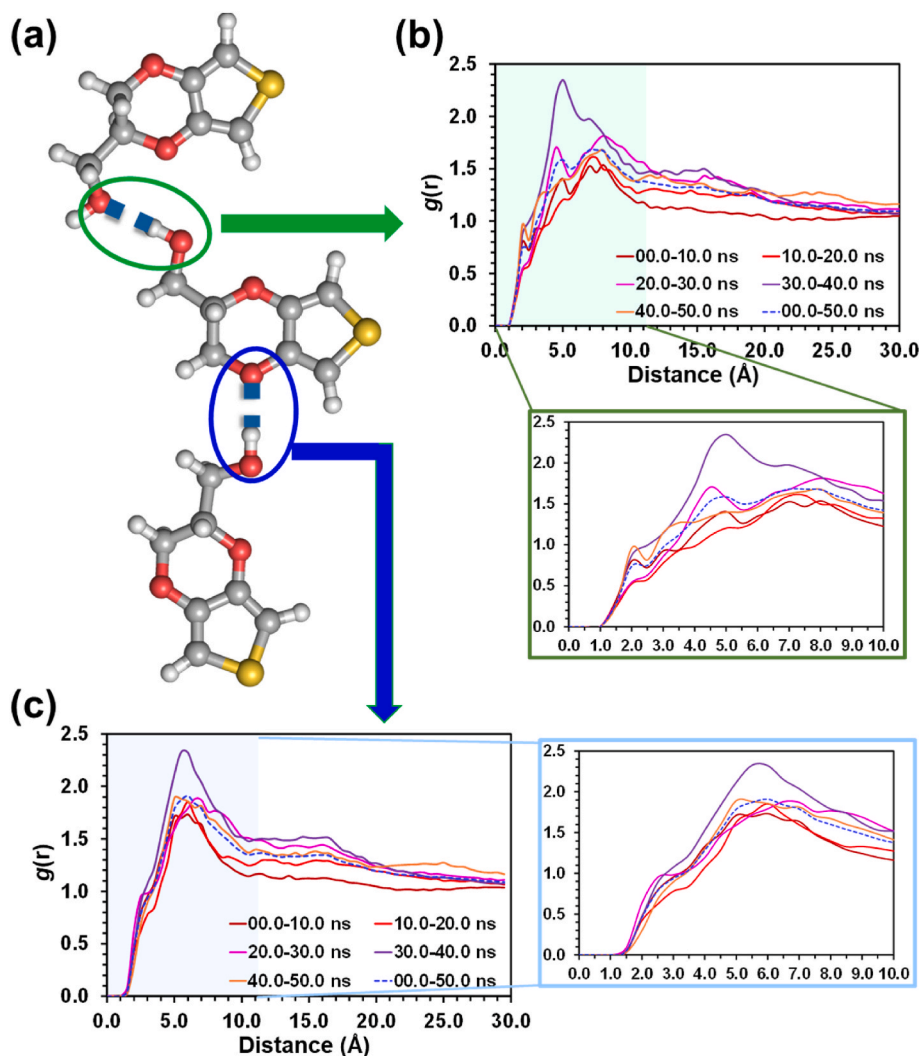


Fig. 7. Analyses of inter-monomer EDOT-MeOH...EDOT-MeOH hydrogen bonding interactions. (a) Sketch showing the O-H...O(H) (green circle) and O-H...O(C) (blue circle) hydrogen bonds. (b, c) EDOT-MeOH...EDOT-MeOH radial distribution functions, $g(r)$, for (b) O-H...O(H) and (c) O-H...O(C) hydrogen bonds considering different intervals of time during the MD trajectory. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

that controlled the aggregation of EDOT-MeOH monomers.

The inter-monomer $g(r)$ for O-H...O(C) hydrogen bonds is depicted in Fig. 7c. As occurred for O-H...O(H) hydrogen bonds, O-H...O(C) hydrogen bonds were found to be much less abundant than π - π stacking interactions. While Fig. 7c shows a weak shoulder at 2.5 Å that starts to appear after 10 ns of trajectory, the intense and clearly defined peak at around 5.5 Å evidences the proximity between the monomers, which is consistent with the π - π staking interactions displayed in Fig. 6.

3.5. Clustering analyses of EDOT-MeOH monomers

Based on the detected intermolecular EDOT-MeOH...EDOT-MeOH interactions, the existence of monomer clusters (i.e. aggregates with 2 or more monomers) was investigated by analyzing the snapshots from the production MD. A monomer was considered as a part of a cluster only if at least one of the EDOT-MeOH...EDOT-MeOH interactions assessed in the previous sub-section was present. More specifically, an O-H...O hydrogen bond was considered as formed when the H...O distance was smaller or equal to 2.5 Å and the $\angle$ OH...O angle was larger or equal to 120°. On the other hand, the π - π stacking of thiophene rings was counted when the distance between the centers of masses of two aromatic rings was smaller or equal to 6.5 Å. Finally, EDOT-MeOH assembly was considered as an associative feature if one monomer (A) interacted with another monomer (B), and monomer B presented interactions with a third monomer (C). Then, all three A, B, and C monomers were considered to belong to the same cluster.

Groups of clusters containing 2, 3 and 4 EDOT-MeOH molecules were detected along the production trajectory. In a first approximation, each group was classified into families according to the maximum residence time ($\tau_{\max}$), which allowed defining a series of residence time (τ) intervals. Thus, τ corresponds to the lifetime of each identified cluster. The results according to this classification, which are shown in Table 2, reflected that for each group, the amount of clusters into each family increases with decreasing τ , while τ increases with the decreasing number of EDOT-MeOH molecules containing in the group. Thus, the average residence time ($\langle\tau\rangle$) decreases from 2.20 ± 1.42 ns (family with 5 clusters) to 0.03 ± 0.01 ns (family with 229 clusters) for the group with 2 EDOT-MeOH molecules, from 0.52 ns (family with 1 cluster) to 0.02 ± 0.01 ns (family with 27 clusters) for the group with 3 EDOT-MeOH molecules, and is 0.01 ns for the only cluster with 4 EDOT-MeOH molecules.

The analyses of the interactions responsible for the stability of the

Table 2

Clustering analysis of the groups formed by EDOT-MeOH molecules, which have been classified according to the residence time.

Group ^a	τ intervals ^b (ns)	# clusters	$\langle\tau\rangle$ (ns) ^c	$\langle\text{HB}\rangle$ ^d	$\langle\pi\pi\rangle$ ^d
2	>1	5	2.20 ± 1.42	0.01 ± 0.01	1.00 ± 0.00
	0.5–0.9	10	0.55 ± 0.06	0.10 ± 0.09	0.92 ± 0.06
	0.1–0.5	98	0.19 ± 0.09	0.07 ± 0.07	0.95 ± 0.05
	<0.1	229	0.03 ± 0.01	0.16 ± 0.30	0.86 ± 0.27
3	0.5–0.9	1	0.52	0.15 ± 0.38	1.90 ± 0.39
	0.1–0.5	5	0.17 ± 0.14	0.20 ± 0.24	1.98 ± 0.25
	<0.1	27	0.02 ± 0.01	0.33 ± 0.54	1.76 ± 0.56
4	<0.1	1	0.01	0.00	3.00

^e Average number of π - π stacking interactions.

^a Number of EDOT-MeOH molecules in the group.

^b Intervals of residence time defined after determining $\tau_{\max}$ for each group.

^c Average residence time.

^d Average number of hydrogen bonds.

clusters of each group indicated that π - π stacking interactions were dominant, which is consistent with the inter-monomer $g(r)$ described in Figs. 6 and 7. Table 2 includes the average number of hydrogen bonds and π - π stacking interactions for each group of aggregates, which are denoted $\langle\text{HB}\rangle$ and $\langle\pi\pi\rangle$, respectively. As it can be seen, the values of $\langle\text{HB}\rangle$ and $\langle\pi\pi\rangle$ for the group with 2 EDOT-MeOH molecules are ≤ 0.16 and higher than ≥ 0.86 , respectively, independently of the τ interval. Similarly, $\langle\text{HB}\rangle$ and $\langle\pi\pi\rangle$ are ≤ 0.33 and higher than ≥ 1.76 , respectively, for the group with 3 EDOT-MeOH molecules, whereas the group with 4 EDOT-MeOH is exclusively stabilized by π - π stacking interactions (i.e. $\langle\pi\pi\rangle = 3.00$). These results suggests that the polymerization of infiltrated EDOT-MeOH to form PEDOT-MeOH nanoparticles nucleates from clusters of monomers stabilized by π - π stacking interactions.

It is worth noting that π - π stacking interactions among the aromatic rings favor the inter-ring charge transfer, affecting the conductivity and electrochemical response and precluding the disruption of any conjugation [54–57]. Indeed, it has been reported that intrinsically high conductivity and electrochemical response of PEDOT are enhanced by promoting solvent-driven π - π coupling of polymer chains [58,59]. The size of π - π stacked connected PEDOT clusters is known to depend on the PEDOT chain length so that by increasing the chain length, the morphology evolves to a more efficient network of π - π stacked network, enhancing the electrochemical response [60].

A second approximation was used to classify the aggregates of each group. This involved the accumulated life time (t_{acc}), which refers to the total time in which a given cluster is formed during the production MD trajectory. Accordingly, after determining the maximum accumulated life time ($t_{\text{acc,max}}$) for each group of clusters, a clustering analyses into families was conducted by defining a series of t_{acc} intervals. Results derived from such criterion are displayed in Table 3.

The group involving 2 EDOT-OH molecules was distributed in six families with a number of clusters ranging from 2 to 235. The average values of the accumulated life time ($\langle t_{\text{acc}} \rangle$), which are listed in Table 3, showed 6 clusters with $t_{\text{acc}} > 5$ ns. More specifically, such clusters remained stable during a significant part of the production trajectory, which ranged from 13 % for $\langle t_{\text{acc}} \rangle = 6.56 \pm 1.22$ ns to 42 % for $\langle t_{\text{acc}} \rangle = 20.93 \pm 0.52$ ns. In all cases, the two monomers interacted exclusively through the π - π stacking of the thiophene rings, being $\langle\pi\pi\rangle \approx 1$ and $\langle\text{HB}\rangle \approx 0$. For the families with $t_{\text{acc}} \leq 5$ ns, both the amount of clusters and $\langle\text{HB}\rangle$ increased with decreasing $\langle t_{\text{acc}} \rangle$, while $\langle\pi\pi\rangle$ decreased. For the group with 3 EDOT-OH molecules, three families with $\langle t_{\text{acc}} \rangle \leq 2.6$ ns were identified. Although the predominant interaction was also the π - π stacking, the contribution of hydrogen bonding was not negligible in that group with $\langle\text{HB}\rangle$ ranging from 0.2 to 0.4. Finally, the unique cluster with 4 EDOT-MeOH molecules, which showed a t_{acc} of only 0.01 ns, involved 3 neat π - π stacking interactions.

Table 3

Clustering analysis of the groups formed by EDOT-MeOH molecules, which have been classified according to the accumulated life time.

Group ^a	t_{acc} ^b (ns)	# clusters	$\langle t_{\text{acc}} \rangle$ (ns) ^c	$\langle\text{HB}\rangle$ ^d	$\langle\pi\pi\rangle$ ^d
2	>20	2	20.93 ± 0.52	0.00	1.00 ± 0.00
	10–20	2	13.41 ± 1.26	<0.01	1.00 ± 0.01
	5–10	2	6.56 ± 1.22	0.04 ± 0.02	0.99 ± 0.01
	1–5	56	2.20 ± 1.04	0.08 ± 0.01	0.94 ± 0.07
	0.5–0.9	47	0.72 ± 0.15	0.10 ± 0.03	0.93 ± 0.10
	<0.5	235	0.16 ± 0.13	0.16 ± 0.09	0.87 ± 0.27
3	1–5	2	2.51 ± 0.69	0.24 ± 0.13	1.87 ± 0.04
	0.5–1	5	0.68 ± 0.08	0.48 ± 0.10	1.67 ± 0.92
	<0.5	26	0.11 ± 0.12	0.28 ± 0.09	1.82 ± 0.46
4	<0.5	1	0.01	0.00	3.00

^e Average number of π - π stacking interactions.

^a Number of EDOT-MeOH molecules in the group.

^b Intervals of accumulated life time defined after determining $t_{\text{acc,max}}$ for each group.

^c Average accumulated life time.

^d Average number of hydrogen bonds.

Overall, results displayed in Table 3 support and complete those listed in Table 2, indicating that the infiltrated EDOT-MeOH monomers tend to interact forming small clusters stabilized by π - π stacking interactions. These small groups of monomers are expected to act as nuclei for the chemical oxidative polymerization and evolve towards well-defined PEDOT-MeOH nanoparticles. This hypothesis, which is empirically supported by the SEM observations, is in line with the rest of suggestions derived from MD simulations on clickHA and clickHA/PEDOT-MeOH hydrogels.

4. Conclusions

This study has contributed to understand the formation and organization of PEDOT-MeOH nanoparticles in clickHA hydrogels, which are obtained by *in situ* chemical oxidative polymerization of infiltrated EDOT-MeOH monomers. In addition to SEM and CV studies on different clickHA/PEDOT-MeOH hydrogels, which have been used to examine the influence of the amount of infiltrated EDOT-MeOH monomer, MD trajectories have been run considering model systems of clickHA and clickHA/EDOT-MeOH. Results have allowed us to ascertain the structural influence, diffusion and distribution of the monomer molecules when they are confined inside the clickHA hydrogel.

The electrochemical activity of clickHA/PEDOT-MeOH hydrogels increases with the concentration of infiltrated monomer, even though the percolation between PEDOT-MeOH chains was already detected for the system with 0.05 M EDOT-MeOH. On the other hand, PEDOT-MeOH chains organize forming small nanoparticles adhered to the walls of the clickHA hydrogel, independently of the EDOT-MeOH concentration. Interestingly, MD simulations showed that the confined EDOT-MeOH molecules do not significantly alter the internal structure of the clickHA hydrogel. Thus, properties like, for example, the density, unoccupied space, SASA and porosity, were very similar for the two modeled systems. On the other hand, the diffusion of the EDOT-MeOH monomer inside the clickHA hydrogel was much slower than that of water molecules. Despite this, the production MD trajectory was large enough for the monomer molecules to fully explore the simulation box and, consequently, have the opportunity to cluster together through stabilizing inter-monomer interactions. In fact, the analysis of the interactions between EDOT-MeOH molecules revealed the formation of small clusters of between 2 and 4 molecules stabilized by π - π stacking interactions. Thus, the π - π stacking interactions between thiophene rings have turned out to be much more abundant than the hydrogen bonds between the hydroxyl groups of neighboring EDOT-MeOH molecules. Indeed, the last groups fundamentally seem to be destined to form hydrogen bonds with water molecules, which explains the affinity of this monomer with water.

In summary, MD simulation results support that infiltrated EDOT-MeOH monomers tend to group forming small clusters, which are stabilized by π - π stacking interactions and act as nuclei for the chemical oxidative polymerization. This allows to explain the organization of semi-interpenetrated PEDOT-MeOH into a homogenous distribution of nanoparticles, as observed by SEM. These results illustrate the potential of synergistic experimental-simulation investigations to provide understanding at the molecular level of micro- and macro-empirical observations.

CCRediT authorship contribution statement

David Zanuy: Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Victor Castrejón-Comas:** Writing – review & editing, Validation, Investigation, Formal analysis. **Joel Sánchez-Morán:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Adrián Fontana-Escartín:** Writing – review & editing, Validation, Investigation, Formal analysis. **Marc Arnau:** Writing – review & editing, Validation, Investigation, Formal analysis. **Maria M. Pérez-Madrigrá:**

Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Formal analysis. **Carlos Alemán:** Writing – original draft, Visualization, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymer.2025.128395>.

Data availability

Data will be made available on request.

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